Fleet-Wide Condition Monitoring via Shared Nominal Representations and Local Radius Calibration

Enabling OEM-Centralized, Fleet-Deployed CBM

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Abstract

Objectives: Transportation operators deploying condition monitoring across large fleets of heterogeneous assets rarely have fault labels, time for per-asset representation learning, or long field calibration campaigns. This paper addresses how a single nominal representation can be deployed fleet-wide on day one, without retraining a model or tuning a threshold for every asset.

Methods: A canonical nominal representation, a Conv1D autoencoder, is trained once offline on simulated healthy signals and transferred unchanged to every asset. Each asset then estimates its operational radius, an empirical residual quantile, from a short stream of unlabeled local observations, without retraining the shared model. The method is evaluated on four Case Western Reserve University (CWRU) bearing conditions, spanning healthy, ball-fault, inner-race, and outer-race windows, against an oracle calibrated on each condition's complete healthy dataset. Three calibration strategies are compared: direct threshold transfer, a complete-data oracle radius, and the proposed online radius estimated from only the first 100 healthy windows. The deployment principle is further validated on four independently monitored bearings from the NASA IMS run-to-failure dataset and compared with autoencoders trained separately on each bearing's early-life data.

Findings: Direct transfer of the absolute simulator threshold produced complete false alarms in the field, confirming that the representation, not its absolute score scale, transfers. Using only a 100-window online calibration, the shared representation reached an average healthy false-alarm rate of 3.09%, ball-fault detection of 99.69%, and 100% detection of inner- and outer-race faults, with the radius converging after 51.0 ± 12.6 windows, approaching the asset-specific reference. On NASA IMS, the shared representation detected persistent departure in all four bearings, on average 83.25 hours before the documented end of the experiment, compared with 47.89 hours for the per-asset early-life autoencoders, while eliminating per-asset representation training.

Novelty: The contribution is a fleet-deployment methodology, not a new anomaly-detection architecture: it separates representation learning, performed once and shared fleet-wide, from decision-boundary calibration, performed independently by each asset.

Practical Applications: A shared representation can be developed and maintained centrally, for example by an original equipment manufacturer, and distributed unchanged across an operator's fleet, while each installed asset requires only seconds of unsupervised local calibration and lightweight edge inference, removing the need for fault labels or per-asset model training during rollout.

Keywords: OEM-centralized deployment, shared nominal representation, simulation-to-field transfer, online operational-radius calibration, edge-based condition monitoring

1 Introduction

Modern transportation systems increasingly rely on condition-monitoring technologies to supervise large populations of operational assets, including rolling-element bearings, axleboxes, traction motors, gearboxes, pumps, actuators, vehicles, and spatially indexed infrastructure segments Jardine et al. 2006; Zhao et al. 2019. Rolling-element bearings are among the most safety-critical of these components: railway axlebox bearings alone are monitored across thousands of independently operating vehicles, where defects can propagate to derailment-level failures if undetected Entezami et al. 2020. Although machine-learning research has produced increasingly capable anomaly detectors Chandola et al. 2009; Zhao et al. 2019, the deployment problem remains largely unresolved: a detector that performs well for one asset under one calibrated operating condition does not automatically scale to hundreds or thousands of independently installed units.

At fleet rollout, several constraints arise simultaneously. First, representative fault examples are rarely available. Second, waiting months or years for failures to accumulate is operationally unacceptable. Third, installation-specific factors such as sensor coupling, operating point, mounting, and environmental noise alter the numerical scale of anomaly scores between assets Randall and Antoni 2011. In railway fleets, for example, axlebox bearings of nominally identical rolling stock are exposed to different track quality, wheel wear, load distribution, and wayside versus on-board sensor placement, so that a detector calibrated on one vehicle rarely transfers its absolute decision threshold to another Entezami et al. 2020. Finally, training, calibrating, validating, and maintaining a separate representation model for every physical unit creates an engineering burden that grows directly with fleet size.

The practical question is therefore not simply

> **Q1.** *How can an anomalous condition be detected?*

but rather

> **Q2.** *How can one nominal representation be deployed on day one across a heterogeneous fleet while retaining performance close to an ideal asset-specific solution, without per-asset representation learning or manual threshold tuning?*

This paper addresses Q2 by separating representation engineering from deployment calibration. A canonical nominal representation is learned once, offline, from simulated healthy observations and distributed unchanged across the fleet. Each deployed asset estimates only its own operational radius from a short stream of unlabeled local observations. The shared model is never retrained, fine-tuned, or adapted in the field.

This separation also defines an OEM-centralized deployment model. The shared nominal representation is engineered, validated, versioned, and maintained centrally, for example by the original equipment manufacturer (OEM), whereas each deployed asset performs only a lightweight local calibration during commissioning. Consequently, fleet expansion no longer requires developing, validating, and maintaining an independent representation model for every physical unit. Instead, deployment consists only of distributing the shared representation and estimating one local operational radius for each asset.

From a broader perspective, this separation can also be interpreted as a local inverse calibration problem. Representation learning is performed once to recover a canonical nominal representation, whereas deployment requires each asset only to identify its own operational radius around that fixed representation. This interpretation follows the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026a; Di Santi 2026b, while the present paper focuses on their engineering realisation for scalable fleet deployment.

The transferable object is therefore not an anomaly threshold but the canonical nominal representation itself. Absolute reconstruction-error thresholds learned in simulation generally do not transfer because their numerical scale depends on the simulator, preprocessing, and residual distribution. Instead, the proposed methodology transfers the shared representation and a fleet-wide nominal coverage, while each asset estimates locally the empirical residual quantile that realizes that coverage.

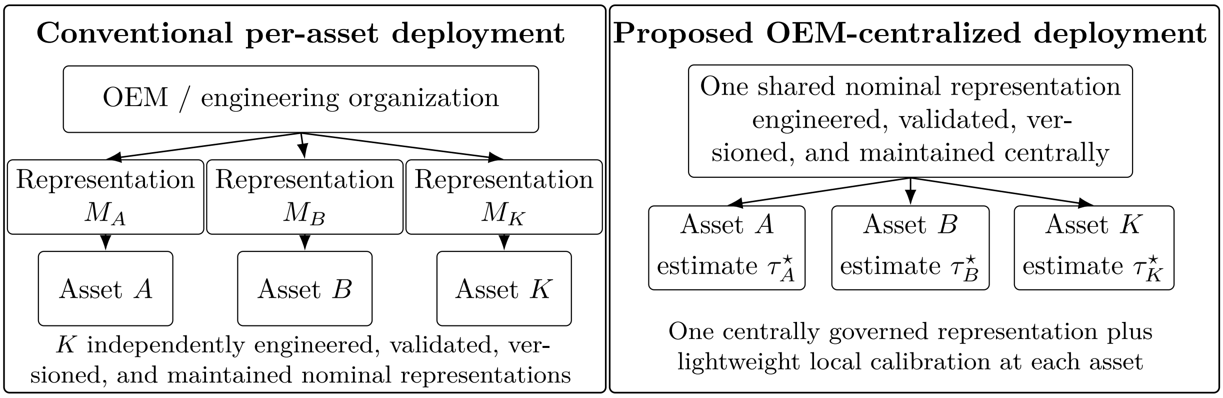

Figure 1: Comparison of conventional per-asset representation learning and the proposed OEM-centralized deployment architecture. Conventional deployment requires independently engineering and maintaining a nominal representation for each monitored unit. The proposed method distributes one shared nominal representation unchanged across the fleet, while each asset estimates only its local operational radius during commissioning.

This formulation also changes the relevant experimental comparison. The ideal reference is not another globally shared detector but an operationally unrealistic situation in which every asset has its complete healthy dataset available for independent representation learning and calibration. Evaluating Q2 requires answering a third question:

> **Q3.** *How much detection performance is lost when many asset-specific models are replaced by one simulator-trained representation and a few seconds of local calibration?*

We answer Q3 first under controlled operating-condition heterogeneity using four CWRU conditions Smith and Randall 2015, and then evaluate its extension to independently monitored physical units undergoing naturally evolving degradation using the NASA IMS run-to-failure dataset.

The asset-specific reference methods are trained and calibrated using the complete healthy dataset of each operating condition. In contrast, the proposed method uses no field observations for representation learning and estimates the local operational radius from only the first 100 healthy windows, corresponding to approximately ten seconds of signal.

The experimental results show that the cost of scalable deployment is small. Asset-specific autoencoders reach 100% detection across the evaluated ball-, inner-race-, and outer-race-fault families. The unchanged simulator-trained shared model, combined with online local-radius estimation, reaches an average ball-fault detection rate of 99.69% and 100% detection of inner- and outer-race faults. The operational radius converges after 51.0 ± 12.6 windows on average. Direct transfer of the absolute simulator threshold, however, produces 100% healthy false alarms, confirming that the representation transfers whereas its absolute score scale does not.

The contribution of this paper is therefore a deployment methodology rather than a new anomaly- detection architecture. Although a Conv1D autoencoder is used as the experimental representation, the deployment principle is representation-agnostic. The principal contributions are:

1. an OEM-centralized fleet-deployment architecture in which a single shared canonical nominal representation is engineered, validated, and maintained centrally, while deployed assets perform only lightweight local operational-radius calibration;

2. a direct simulation-to-field deployment in which the representation is transferred unchanged without field-data retraining;

3. an online empirical-quantile calibration procedure that estimates an operational radius from a short unlabeled warm-up;

4. a convergence criterion separating finite cold-start calibration from subsequent degradation monitoring;
5. an evaluation against ideal asset-specific detectors trained and calibrated with complete healthy datasets; and
6. an edge-oriented deployment model that eliminates per-asset representation learning after installation, allowing fleet-wide deployment through distribution of one shared representation and local unsupervised calibration.

The remainder of the paper is organized as follows. Section 2 reviews condition monitoring, one-class anomaly detection, adaptive calibration, and fleet deployment. Section 3 introduces the deployment methodology. Section 4 presents the experimental protocol. Section 5 reports the deployment results. Section 6 summarizes the mathematical interpretation underlying the deployment guarantees. Section 7 closes with a discussion of industrial implications, limitations, and future work.

2 Related Work

Condition-based maintenance (CBM) has become a central component of modern transportation asset management, with condition monitoring providing the operational evidence required to supervise large fleets of heterogeneous assets Jardine et al. 2006; Chandola et al. 2009; Zhao et al. 2019. Although sensing modalities differ across applications, the computational objective remains the same: to characterize nominal operation and detect persistent departures that may indicate degradation.

Most machine-learning approaches formulate this problem as supervised fault classification, requiring representative fault examples during training. While effective on benchmark datasets, these methods assume that deployment conditions remain close to those observed during model development.

One-class methods, including one-class SVM Schölkopf et al. 1999, neural-network novelty detection Bishop 1994, and autoencoder-based anomaly detection Hinton and Salakhutdinov 2006; Sakurada and Yairi 2014, reduce the dependence on fault labels by learning only nominal behaviour. More recent representation-learning approaches further improve anomaly detection by constructing latent spaces in which healthy operation occupies a compact region while degraded behaviour becomes more easily separable.

However, most existing work focuses on improving anomaly detectors rather than addressing the deployment problem itself: how a single nominal representation can be initialized, calibrated, and maintained across heterogeneous fleets.

Adaptive thresholds have been widely used to compensate for changing operating conditions. However, unrestricted adaptation risks absorbing progressive degradation into the nominal baseline. This motivates separating calibration from monitoring. In the proposed methodology, adaptation is limited to a finite cold-start phase during which each asset estimates its local operational radius. Once convergence is achieved, the shared representation remains fixed and subsequent departures become diagnostic evidence rather than input for further adaptation.

The proposed deployment strategy is also related to invariant representation learning Bronstein et al. 2021 and anomaly detection under distribution shift Carvalho et al. 2023, both of which seek representations that are insensitive to nuisance variability. Our contribution, however, lies at a different level. Rather than proposing a new representation-learning architecture, we address the deployment problem by combining a shared nominal representation with local operational-radius calibration, eliminating per-asset representation learning while preserving a common nominal coverage across the fleet.

A related and rapidly growing literature addresses cross-condition and cross-machine transfer through domain adaptation, typically aligning source and target feature distributions via maximum mean discrepancy, adversarial domain classifiers, or pseudo-labeled fine-tuning Wang et al. 2026; Hue et al. 2025; Chen et al. 2026. These methods substantially reduce the need for labeled target-domain faults and have demonstrated strong cross-condition accuracy on CWRU and related benchmarks. However, they still require access to target-domain observations during an explicit adaptation or fine-tuning stage, executed independently for each new operating condition or physical unit. This reintroduces, at a smaller scale, the same per-asset engineering burden that fleet-wide deployment seeks to eliminate: every newly installed asset still triggers a training procedure, even when it requires no fault labels. In contrast, the proposed methodology performs no adaptation, fine-tuning, or parameter update on the shared representation at deployment time; each asset only estimates a scalar operational radius from unlabeled observations, without any gradient-based training step. The distinction is therefore not between labeled and unlabeled deployment, but between deployment procedures that retrain or adapt a model per asset and one that requires no training step whatsoever after the representation is fixed.

This viewpoint is also consistent with the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026b; Di Santi 2026a, although the present paper focuses exclusively on their engineering realization for scalable fleet deployment.

3 Method

This section formalizes the proposed deployment methodology, which separates a fleet-wide shared nominal representation from asset-specific operational-radius calibration.

3.1 Deployment architecture: shared representation, local radius

Let $x_{k,t} \in \mathcal{X}$ denote an observation window acquired from operational unit k at time t. In the bearing experiments, $x_{k,t}$ is instantiated as a fixed-length vibration window. A shared offline-trained representation produces an anomaly score

$$r_{k,t} = r(x_{k,t}). \tag{1}$$

The representation is shared across the fleet, whereas the decision boundary is estimated independently for each asset. Operational unit k maintains a local operational radius $\tau_k(t)$ and raises an alarm whenever

$$r_{k,t} > \tau_k(t). \tag{2}$$

The deployment methodology therefore follows three principles:

1. one shared representation for the fleet;
2. local calibration for each asset; and
3. no per-asset representation retraining.

The representation is shared across the fleet, whereas the decision boundary is estimated independently for each asset. This separation matches the OEM-centralized deployment model introduced in Section 1: Eq. (1) is engineered, validated, and versioned once by whichever party owns the representation, for example the component OEM, a third-party analytics provider, or the operator's own engineering organization, while Eq. (2) is evaluated independently and automatically by each deployed asset. Operational unit k maintains a local operational radius $\tau_k(t)$ and raises an alarm whenever

3.2 Representation instantiation

The proposed deployment methodology is instantiated with a Conv1D autoencoder $A_\phi = D_\phi \circ E_\phi$ trained exclusively on nominal signals. The encoder maps a 1200-sample window through three Conv1D layers (32, 64, and 128 filters) to a 16-dimensional latent code with an ℓ_1 activity penalty, while the decoder mirrors this architecture using Conv1DTranspose layers. The network contains 847,601 trainable parameters and is optimized with Adam, mean-squared reconstruction loss, early stopping, and learning-rate decay.

The anomaly score is the reconstruction residual

$$r(x) = \|x - A_\phi(x)\|_2^2. \tag{3}$$

The contribution of this paper lies in the deployment methodology rather than in a new representation-learning architecture. The Conv1D autoencoder serves as the nominal representation throughout the experimental validation, while the deployment principles introduced here are independent of this particular training procedure.

3.3 Shared nominal coverage and local empirical radius

The fleet-wide calibration parameter is a target nominal coverage level $1 - \alpha$, rather than an absolute anomaly-score threshold. In the present experiments, $1 - \alpha = 0.98$.

For operational asset k, let $\widehat{Q}_{k,t}(1-\alpha)$ denote the empirical $(1-\alpha)$-quantile of the local residuals observed during initialization up to time t. The local operational radius is

$$\tau_k(t) = \widehat{Q}_{k,t}(1-\alpha). \tag{4}$$

The simulator threshold

$$\tau_{\mathrm{MC}} = \widehat{Q}_{\mathrm{MC}}(1-\alpha) \tag{5}$$

is retained only for comparison. Its absolute numerical value is not used by the proposed field-deployment procedure. What is shared across the fleet is the target nominal coverage, while each asset identifies the residual value that realizes that coverage locally.

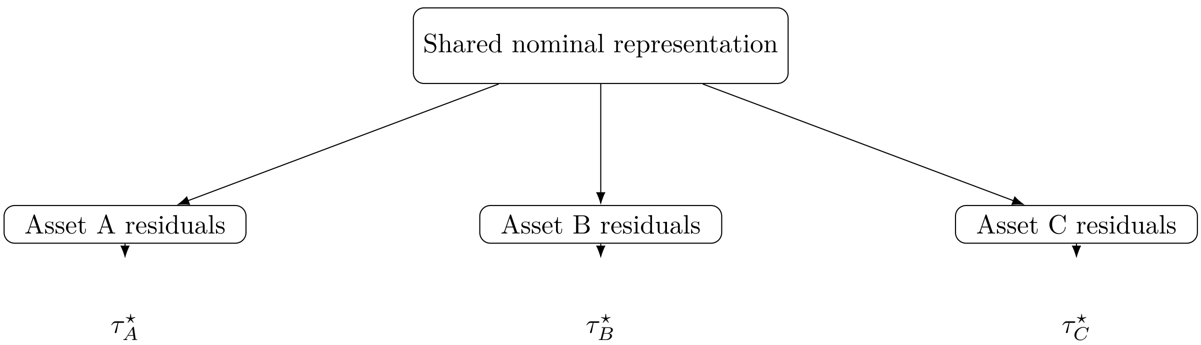

Figure 2: The same shared nominal representation produces reconstruction residuals for every deployed asset. Only the local empirical residual distribution changes across installations. Each asset independently estimates its own operational radius $\tau_k^\star$, while the representation remains unchanged.

3.4 Operational-radius convergence

During cold start, each asset updates Eq. 4 using the incoming nominal residuals. Calibration is declared when the operational radius stabilizes.

For a look-back interval L, define the convergence statistic

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t-L)|}{\max\left(\tau_k(t-L), \epsilon\right)}, \tag{6}$$

where $\epsilon > 0$ prevents division by zero.

Calibration is declared when

$$C_k(t) < \varepsilon_C \tag{7}$$

for m consecutive updates. The converged operational radius is then

$$\tau_k^\star = \tau_k(t_{\mathrm{conv}}). \tag{8}$$

The key hypothesis evaluated in this paper is: *Under stable nominal operation, every asset converges rapidly to its own operational radius around a fixed shared representation.*

After convergence, $\tau_k^\star$ is frozen (or only robustly corrected) to prevent progressive degradation from being absorbed into the nominal baseline. The convergence criterion therefore marks the transition from calibration to operational monitoring.

3.5 Guarded updates

For deployments requiring continued adaptation beyond the controlled warm-up, a guarded update rule can be applied to the local baseline. Let $\delta > 0$ define a guard band. Local statistics are updated only when

$$r_{k,t} < \tau_k(t) - \delta. \tag{9}$$

The three operating regions are

$$r_{k,t} < \tau_k(t) - \delta \quad \Rightarrow \quad \text{safe update,} \tag{10}$$

$$\tau_k(t) - \delta \le r_{k,t} \le \tau_k(t) \quad \Rightarrow \quad \text{observe, do not update,} \tag{11}$$

$$r_{k,t} > \tau_k(t) \quad \Rightarrow \quad \text{anomaly.} \tag{12}$$

This rule requires only a residual buffer (or running robust statistics), the current operational radius, and the convergence state.

3.6 Deployment states

Each asset progresses through three deployment states:

1. **Cold start:** local estimation of the operational radius;
2. **Converged monitoring:** $\tau_k^\star$ is fixed and anomaly detection is active;
3. **Operational departure:** persistent exceedances of the converged operational radius.

In practice, alarms may require q exceedances within the last p observations to avoid reacting to isolated impulses.

3.7 Scope of the Method

The proposed methodology is independent of asset type and sensing modality. Throughout the paper, the index k denotes an independently monitored operational unit, such as a bearing, traction motor, gearbox, pump, actuator, vehicle, or infrastructure segment.

The methodology requires only (i) a shared nominal representation, (ii) repeated observations from each operational unit, and (iii) local estimation of the operational radius.

The CWRU bearing benchmark is used solely as a controlled validation domain.

4 Experimental Design

The experimental protocol evaluates the proposed deployment methodology rather than the underlying representation-learning architecture. Across the five experimental stages, we measure:

- false-alarm rate during cold start;
- number of accepted samples to operational-radius convergence;
- variability of $\tau_k^\star$ across assets and regimes;
- post-convergence detection rate;
- sensitivity to buffer size and convergence parameters;
- performance of absolute-threshold transfer versus shared-coverage calibration;
- cost of per-asset retraining avoided by the shared-model design.

Stage 1 — Controlled multi-asset simulation. Four bearing streams are generated with a physics-motivated nominal simulator using 1200-sample windows, a 30 Hz shaft frequency, and bounded variation in amplitude, phase, speed, offset, and noise. The shared model is trained on 4,000 simulated nominal windows, with 1,000 additional nominal windows reserved for validation. Faults are introduced only during evaluation. A separate synthetic test set contains 750 windows per regime. Each simulated asset begins in nominal operation and independently calibrates its radius. Asset A remains nominal; Assets B–D transition after convergence to outer-race, inner-race, and ball-fault regimes.

This stage measures: (i) samples to convergence; (ii) false alarms before and after convergence; (iii) the separation between $r_{k,t}$ and $\tau_k^\star$ after fault onset; and (iv) whether each nominal asset reaches a stable but asset-specific radius.

Stage 2 — Direct simulation-to-field deployment on CWRU. The nominal representation trained in Stage 1 is transferred unchanged to the Case Western Reserve University (CWRU) Drive-End dataset. No CWRU observation is used for representation learning, retraining, or fine tuning.

Each operating condition is treated as an independent deployment environment. The first 100 healthy windows are used only to estimate the local 98th-percentile operational radius, after which fault observations are evaluated using the unchanged shared model.

This stage evaluates whether (i) the learned representation transfers directly from simulation to field, and (ii) local empirical calibration compensates for the non-transferability of the absolute simulator threshold.

The four CWRU operating conditions provide a controlled evaluation of score-scale heterogeneity and local calibration rather than four physically distinct fleet assets. Validation on independent run-to-failure bearings is addressed separately in Stage 4 using the NASA IMS dataset.

Stage 3 — Reference comparison and calibration ablation. To quantify the cost of scalability, we construct an intentionally favorable but operationally non-scalable reference condition. For each CWRU operating condition, three detectors are trained and calibrated independently using the complete available healthy dataset: (i) a Conv1D autoencoder; (ii) a one-class SVM operating on PCA-reduced standardized windows; and (iii) an Elliptic Envelope operating on PCA-reduced standardized windows.

Each detector uses the empirical 98th percentile of its own healthy anomaly-score distribution as the decision boundary. Fault observations are never used for model fitting or threshold selection. These experiments represent an operationally unrealistic upper bound in which complete nominal data and asset-specific model development are available for every monitored unit.

Using the unchanged simulator-trained autoencoder, we compare three deployment strategies:

1. **Direct threshold transfer:** apply the absolute synthetic p98 residual threshold unchanged to each CWRU condition;

2. **Field oracle:** estimate the p98 operational radius using the complete healthy field dataset of each condition, without retraining the shared representation; and

3. **Online local convergence:** estimate the p98 operational radius progressively from only the first 100 unlabeled healthy windows.

This ablation separates representation transfer from decision-boundary transfer. The representation is identical in all three conditions; only the information available for local calibration changes.

Stage 4 — Multi-unit run-to-failure validation on NASA IMS. A second nominal representation is constructed for the IMS bearing family using simulated healthy observations matched to the dataset sampling and operating characteristics. This model is trained once and then transferred unchanged to the independently monitored bearings within the IMS experiments. Each bearing estimates its own operational radius from an early-life unlabeled segment and subsequently enters post-convergence monitoring.

This stage evaluates whether the proposed deployment principle extends from heterogeneous CWRU operating conditions to physically distinct bearing units and naturally evolving degradation trajectories. We report: (i) samples to operational-radius convergence; (ii) variability of the converged radius across bearings; (iii) healthy-period false-alarm rate; (iv) first persistent departure from the frozen radius; and (v) alarm lead time relative to the documented end of the experiment.

Stage 5 — Edge cost. We report model size, parameter count, inference latency per 1200-sample window on CPU, local state memory, and the reduction in transmitted data when only scores and selected anomalous windows are sent rather than the continuous raw signal.

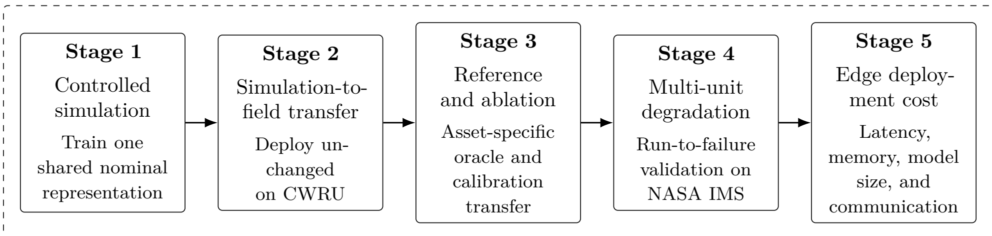

Figure 3: Deployment-oriented experimental protocol. The experiments progress from controlled representation learning to direct field transfer, ideal-reference comparison, multi-unit run-to-failure validation, and measurement of the edge-deployment footprint. Across deployment stages, the nominal representation remains fixed while each monitored unit estimates its own local operational radius.

5 Results

The experiments address three complementary deployment questions. First, the CWRU experiments establish the performance attainable under an ideal but operationally non-scalable condition in which separate detectors are trained and calibrated for each operating condition.

Second, they evaluate whether a simulator-trained shared representation, combined only with short local calibration, can approach that ideal performance without field-model retraining. Third, the NASA IMS run-to-failure experiment evaluates whether the same deployment principle extends to independently monitored physical units undergoing naturally evolving degradation.

The CWRU evaluation includes four operating conditions corresponding to 1730, 1750, 1772, and 1797 RPM. Across these conditions, the experiments contain 1,414 healthy windows, 1,621 ball-fault windows, 1,621 inner-race-fault windows, and 2,846 outer-race-fault windows. All windows contain 1,200 samples and are processed identically across detectors. The NASA IMS evaluation subsequently considers four independently monitored bearings from Experiment 2, each represented by 984 chronologically ordered recordings, of which the first 100 are reserved for early-life calibration.

5.1 Ideal asset-specific reference

Table 1 reports the performance of three detectors trained independently for each CWRU operating condition using the complete healthy dataset available for that condition. These experiments constitute an oracle reference rather than a deployable fleet architecture, assuming complete nominal data and independent training and calibration for every operating condition.

Since each detector is calibrated at the empirical 98th percentile of its healthy anomaly-score distribution, the healthy false-alarm rate is approximately 2% by construction. The comparison therefore evaluates the diagnostic quality of the anomaly score rather than threshold estimation.

Table 1: Ideal asset-specific detector performance on CWRU. Values are mean ± standard deviation across the four operating conditions. Each detector is trained and calibrated separately using the complete healthy dataset of the corresponding condition.

Detector	Healthy FAR (%)	Ball DR (%)	Inner-race DR (%)	Outer-race DR (%)
Asset-specific AE	2.29 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
Asset-specific OC-SVM	2.29 ± 0.12	3.09 ± 6.17	25.05 ± 0.13	2.71 ± 3.15
Asset-specific Elliptic Envelope	2.29 ± 0.12	0.62 ± 1.23	14.43 ± 10.63	1.76 ± 3.15

The asset-specific autoencoder achieves 100% detection for all evaluated fault families under all four operating conditions. In contrast, the classical one-class detectors reproduce the target false-alarm rate but provide little diagnostic separation, with average ball-fault detection rates of 3.09% (OC-SVM) and 0.62% (Elliptic Envelope).

These results indicate that nominal calibration alone is insufficient. Although all detectors achieve the prescribed healthy false-alarm rate, only the reconstruction-based representation consistently separates healthy and faulty operation. The asset-specific autoencoder is therefore used as the oracle reference for evaluating the cost of shared-model deployment.

Repeating the complete training and evaluation procedure five times under independent random initializations produced identical healthy false-alarm and fault-detection performance across all operating conditions; only the absolute reconstruction threshold varied slightly between runs (0.459–0.759), without affecting diagnostic performance. This confirms that stochastic optimization converges to diagnostically equivalent nominal representations despite differing absolute residual scales.

5.2 Direct simulation-to-field deployment

The shared autoencoder is trained once using simulated healthy observations and is transferred unchanged to all four CWRU conditions. No CWRU observation is used to retrain, fine-tune, or otherwise modify the model or its preprocessing transformation.

Three calibration conditions are evaluated:

1. **Direct threshold transfer**, in which the absolute p98 residual threshold estimated in simulation is applied unchanged in the field;
2. **Field oracle**, in which the p98 operational radius is computed using the complete healthy field dataset for each condition; and
3. **Proposed online convergence**, in which the local p98 operational radius is estimated progressively from only the first 100 unlabeled healthy windows.

Table 2 reports the complete results.

Table 2: Simulation-to-field deployment of the shared autoencoder on CWRU. FAR denotes the healthy false-alarm rate and DR the fault detection rate. The field-oracle radius uses every available healthy window; the online radius uses only the first 100 healthy windows. A dash indicates that convergence is not applicable because the threshold is computed directly rather than estimated sequentially.

RPM	Calibration	τ	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Healthy windows	Warm-up (windows)	Convergence (windows)
1730	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	404	0	–
1730	Field oracle	0.2005	2.23	100.00	100.00	100.00	404	404	–
1730	Proposed online convergence	0.1984	2.97	100.00	100.00	100.00	404	100	38
1750	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	404	0	–
1750	Field oracle	0.1786	2.23	100.00	100.00	100.00	404	404	–
1750	Proposed online convergence	0.1759	2.97	100.00	100.00	100.00	404	100	58
1772	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	403	0	–
1772	Field oracle	0.1997	2.23	100.00	100.00	100.00	403	403	–
1772	Proposed online convergence	0.2072	0.50	100.00	100.00	100.00	403	100	43
1797	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	203	0	–
1797	Field oracle	0.2474	2.46	96.80	100.00	100.00	203	203	–
1797	Proposed online convergence	0.2364	5.91	98.77	100.00	100.00	203	100	65

Direct transfer of the absolute simulator threshold fails under every operating condition. Its 100% apparent fault-detection rate is meaningless because it simultaneously classifies 100% of the healthy field windows as anomalous. The shared representation therefore transfers, but its absolute reconstruction-error scale does not.

The field-oracle calibration restores the target false-alarm level, obtaining an average healthy FAR of $2.29 \pm 0.12\%$. With this complete knowledge of each field condition, the unchanged simulator-trained model obtains $99.20 \pm 1.60\%$ ball-fault detection and 100% detection of both inner- and outer-race faults.

The proposed online procedure reaches comparable performance using only the first 100 healthy windows. Across the four operating conditions, it achieves an average healthy FAR of $3.09 \pm 2.22\%$, ball-fault detection of $99.69 \pm 0.62\%$, and 100% detection of both inner- and outer-race faults. The online operational radius converges after 51.0 ± 12.6 windows, with observed convergence indices ranging from 38 to 65 windows.

5.3 Cost of scalable deployment

Table 3 compares the proposed shared deployment with an ideal but operationally non-scalable asset-specific reference. The comparison intentionally favors the baseline: the oracle detector is trained and calibrated using the complete healthy dataset of each operating condition, whereas the proposed method estimates its local operational radius from only the first 100 healthy windows and performs no field-model retraining.

Table 3: Ideal but non-scalable asset-specific deployment versus the proposed fleet-scalable deployment. Values are mean ± standard deviation across the four CWRU operating conditions.

Deployment strategy	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Field model training	Healthy field data used
Asset-specific AE oracle	2.29 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	One model per condition	Complete dataset
Shared AE with field-oracle radius	2.29 ± 0.12	99.20 ± 1.60	100.00 ± 0.00	100.00 ± 0.00	None	Complete dataset
Proposed shared AE with online radius	3.09 ± 2.22	99.69 ± 0.62	100.00 ± 0.00	100.00 ± 0.00	None	100-window warm-up

Replacing four independently trained autoencoders with one simulator-trained shared model produces no observed loss for inner- or outer-race detection and less than one percentage point of average difference for ball-fault detection. The principal cost appears in healthy false alarms, whose mean increases from 2.29% to 3.09%. Most of this increase is concentrated in the 1797 RPM condition (5.91%), whereas the remaining conditions range between 0.50% and 2.97%.

These results support the central deployment claim of this paper. A single simulator-trained canonical representation, combined with short local operational-radius calibration, achieves near-oracle detection performance without asset-specific representation learning, substantially reducing deployment complexity.

5.4 Operational-radius convergence

Figure 4 shows the sequential estimation of the local operational radius under the fixed shared representation. For each CWRU operating condition,

$$\tau_k(t) = \mu_k(t) + z_{\mathrm{MC}}\sigma_k(t), \tag{13}$$

where z_{MC} is transferred from the simulated nominal residual distribution. The representation and standardized coverage remain fixed; only the local residual statistics are estimated from incoming healthy observations.

The first 100 windows constitute the prescribed cold-start calibration interval. Across all four operating conditions, the convergence criterion is satisfied between windows 38 and 65, before the calibration budget is exhausted. The operational radii at window 100 are 0.2393, 0.2188, 0.2540, and 0.2777 for 1730, 1750, 1772, and 1797 RPM, respectively.

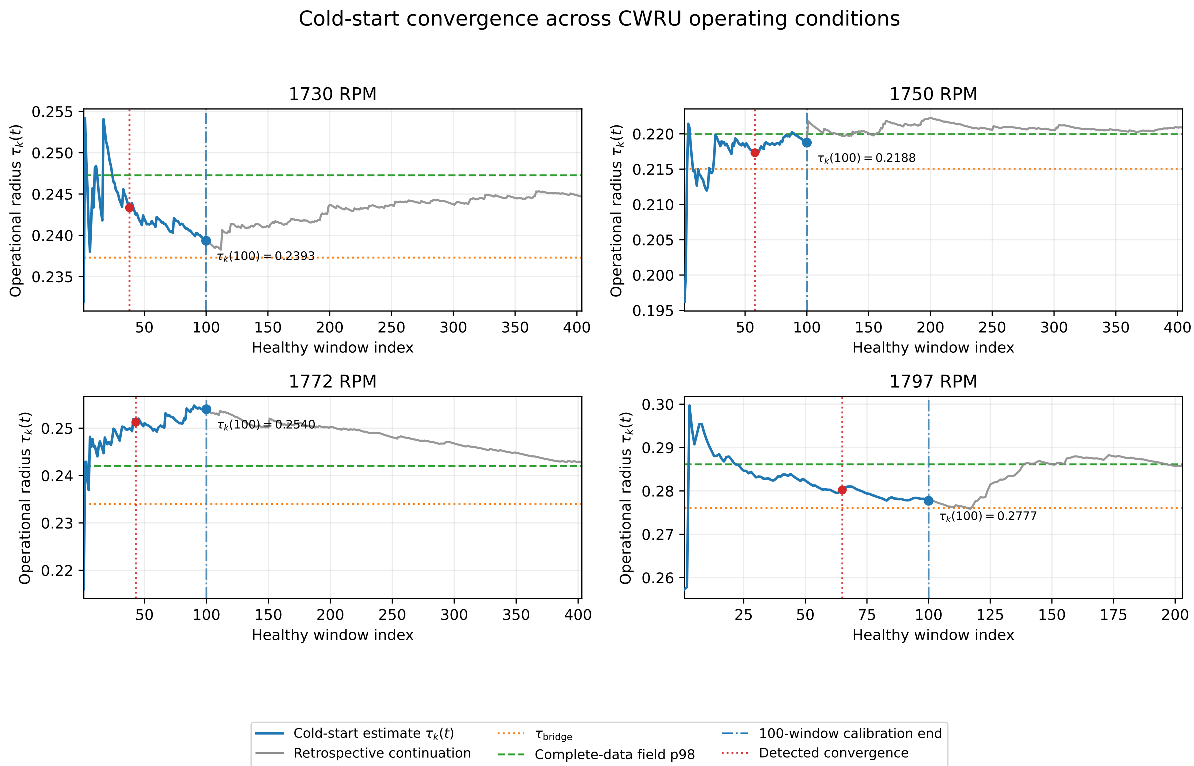

Figure 4: Cold-start convergence and retrospective evolution of the local operational radius across the four CWRU operating conditions. The blue segment shows the sequential estimate during the 100-window calibration interval; the red dotted line marks the detected convergence index. The gray segment is retrospective only and is not used by the deployed detector. Horizontal lines indicate τ_{bridge} and the complete-data field p98 reference.

The red dotted line in Figure 4 marks the detected convergence index, and the blue dash-dotted line marks the end of the prescribed 100-window cold-start interval. The gray segment extends the estimate retrospectively by continuing to incorporate the remaining known-healthy observations; it characterizes the subsequent evolution of the estimator only and is not used to update the deployed decision boundary. The observed post-calibration fluctuations are expected because the healthy residual distribution is not perfectly stationary. Later nominal observations modify the estimated residual mean, variance, or both, causing the cumulative estimate to increase or decrease without implying degradation. This behaviour motivates separating finite calibration from subsequent monitoring.

Operationally, the result demonstrates that an asset-specific acceptance boundary can be established before the complete healthy distribution is available. The complete-data p98 reference assumes prior access to all healthy observations, whereas the proposed procedure reaches a usable local radius during a short initialization interval in every evaluated operating condition.

The present experiment assumes that the initialization data are representative of nominal operation. Detecting assets that are already degraded at installation, or determining that no stable nominal radius can be established, remains outside the present scope and constitutes future work.

5.5 Robustness to local initialization

A practical deployment question is whether local calibration depends critically on the particular healthy observations available during installation. To evaluate this sensitivity, the shared simulator-trained representation, preprocessing, and all model parameters were kept fixed. Only the initialization observations were varied.

For each CWRU operating condition, five independent warm-up sequences were generated by randomly sampling 100 healthy windows with different random seeds. Each sequence independently estimated the local operational radius, after which fault detection was evaluated using the unchanged shared representation.

Table 4: Robustness of online operational-radius estimation under different random warm-up samples. The shared simulator-trained representation remained fixed throughout the experiment. Only the 100 healthy initialization windows used for local calibration were randomly changed. Values are mean ± standard deviation over five random seeds.

RPM	τ_{online}	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Convergence (windows)
1730	0.2905 ± 0.0041	4.08 ± 2.04	99.75 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	45.6 ± 13.2
1750	0.3080 ± 0.0063	3.49 ± 2.47	99.95 ± 0.11	100.00 ± 0.00	100.00 ± 0.00	54.2 ± 13.6
1772	0.3097 ± 0.0076	2.18 ± 1.66	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	49.4 ± 13.8
1797	0.3415 ± 0.0051	3.49 ± 2.44	96.16 ± 0.22	100.00 ± 0.00	100.00 ± 0.00	56.2 ± 15.0

Across all repetitions, the estimated operational radii remained close to their complete-data field-oracle values despite the different initialization samples. Detection performance was correspondingly stable: ball-fault detection remained between approximately 96% and 100%, while inner- and outer-race faults continued to be detected with essentially unchanged performance. The principal variability appeared in the healthy false-alarm rate, which depends on the particular empirical percentile estimated from only 100 initialization windows.

This behaviour is expected statistically because the empirical 98th-percentile is determined by only a few extreme observations. Small changes in those observations therefore produce modest changes in the estimated operational radius without materially affecting diagnostic separation.

These results show that the transferable object remains the shared canonical representation, whereas the local operational radius is consistently recovered from short unlabeled initialization data despite variations in the specific warm-up observations.

5.6 Multi-unit run-to-failure validation on NASA IMS

The CWRU experiments validate simulation-to-field transfer under controlled operating conditions. To evaluate whether the same deployment principle extends to independent physical units undergoing naturally evolving degradation, the shared nominal representation was further evaluated on the NASA IMS run-to-failure dataset.

Each NASA bearing is treated as an independent monitored asset. The simulator-trained autoencoder is transferred unchanged, and no NASA observations are used for representation learning, retraining, or parameter adaptation. Only a short early-life nominal segment is used to estimate the local operational radius.

Table 5 summarizes the recording-level results. The calibration exceedance rate is 2% by construction because the operational radius is defined as the empirical 98th percentile of the early-life scores. After calibration, a persistent departure is detected for all four bearings. The mean monitoring exceedance rate is 28.51%, and the first persistent departure occurs, on average, 83.25 hours before the documented end of the experiment.

Table 5: Recording-level run-to-failure validation on NASA IMS using the shared nominal representation and local operational-radius calibration. Lead time is measured relative to the documented end of the experiment.

Bearing	Calibration exceedance (%)	Persistent departure	Monitoring exceedance (%)	Lead time (h)
1	2.00	Yes	51.47	137.83
2	2.00	Yes	16.86	46.83
3	2.00	Yes	10.29	15.50
4	2.00	Yes	35.41	132.83
Mean	2.00	4/4	28.51	83.25

These results show that the proposed deployment methodology extends beyond controlled operating-condition transfer. A single shared nominal representation can be deployed across independently monitored physical units, requiring only local operational-radius calibration and no per-asset representation retraining.

5.6.1 Stability of local operational-radius estimation

A key deployment requirement is that the local calibration procedure should not depend critically on the particular nominal observations collected during installation. Therefore, the shared representation was kept fixed while multiple random initialization samples were used to estimate the local operational radius.

For each NASA bearing, five independent 100-window samples were drawn from the early-life calibration recordings. The shared representation and its preprocessing remained fixed; only the residual observations used to estimate the local operational radius were varied. Table 6 reports the resulting variability. The estimated radius remained stable across initialization samples, with low relative error and convergence within a limited number of calibration windows.

Table 6: Stability of local operational-radius estimation for the shared NASA representation under different warm-up initializations.

Bearing	τ mean	τ std	Relative error (%)	Convergence (windows)
1	0.0120	0.00013	0.92	48.4
2	0.0165	0.00010	2.07	39.0
3	0.0204	0.00050	2.08	62.2
4	0.0074	0.00009	1.82	45.6
Mean	0.0141	–	1.72	48.8

The stability experiment supports the central deployment hypothesis: the transferable object is the nominal representation, while each physical asset independently identifies its own local acceptance region. Variability between assets is therefore handled by calibration of the operational radius rather than by training separate representations.

5.6.2 Comparison with per-asset early-life training

To quantify the cost of eliminating per-asset representation learning, the proposed deployment was compared with an intentionally favorable reference. For each NASA bearing, an independent autoencoder and preprocessing transformation were trained using its first 100 recordings, repeating the training with five random initializations. The proposed method instead used one frozen simulator-trained representation for all bearings and estimated only a local operational radius.

Table 7 summarizes the recording-level comparison. Both approaches detect a persistent departure for every bearing and reproduce the prescribed 2% calibration exceedance. Monitoring exceedance rates are also similar. Under the present protocol, the shared representation detects the persistent departure earlier for Bearings 1, 2, and 4, while both approaches produce nearly identical timing for Bearing 3. On average, the shared representation increases the lead time from 47.89 h to 83.25 h, providing approximately 35 h of additional warning without requiring asset-specific representation learning.

Table 7: NASA IMS comparison between the proposed shared representation and autoencoders trained separately on the early-life data of each bearing. Asset-specific results are mean ± standard deviation over five training initializations. Shared-model values correspond to one fixed representation applied unchanged to every bearing. Lead time is measured relative to the documented end of the experiment.

Bearing	Deployment strategy	Departure detected (%)	Monitoring exceedance (%)	Lead time (h)
1	Asset-specific early-life AE	100.00 ± 0.00	53.08 ± 1.08	79.90 ± 6.95
1	Shared AE with local radius	100.00	51.47	137.83
2	Asset-specific early-life AE	100.00 ± 0.00	17.69 ± 2.56	33.03 ± 10.80
2	Shared AE with local radius	100.00	16.86	46.83
3	Asset-specific early-life AE	100.00 ± 0.00	10.63 ± 0.00	15.00 ± 0.00
3	Shared AE with local radius	100.00	10.29	15.50
4	Asset-specific early-life AE	100.00 ± 0.00	39.37 ± 1.26	63.63 ± 5.89
4	Shared AE with local radius	100.00	35.41	132.83
Mean	Asset-specific early-life AE	100.00	30.19	47.89
Mean	Shared AE with local radius	100.00	28.51	83.25

The results do not establish universal prognostic superiority across all degradation scenarios. They do demonstrate, however, that eliminating per-asset representation learning incurred no observed loss of detection performance in this experiment. All four bearings reached 100% departure detection, monitoring exceedance rates remained comparable, and the shared representation provided an earlier warning on average (83.25 h versus 47.89 h before the documented end of the experiment), corresponding to an average improvement of approximately 35 h in available lead time.

From a deployment perspective, the difference is substantial. The asset-specific reference requires fitting and maintaining an independent scaler and neural representation for every monitored unit, whereas the proposed methodology requires one centrally maintained representation and only lightweight local calibration at deployment.

5.7 Edge deployment implications

The shared model contains 847,601 trainable parameters and is trained entirely offline. Deployment at the monitored asset requires only forward inference, a small residual buffer, estimation of an empirical quantile, and the current convergence state. No gradient computation, model optimization, fault labels, or asset-specific representation learning are required after installation.

On the evaluation platform, inference over a 1,200-sample window requires a median CPU latency of only 6.26 ms (95th percentile: 6.62 ms), substantially below the 100 ms acquisition interval corresponding to the evaluated windows. Consequently, the proposed deployment operates comfortably in real time while leaving ample computational margin for additional edge tasks such as communication, diagnostics, or signal preprocessing.

The local deployment state is also extremely compact. Besides the shared model itself, each asset stores only a buffer of the most recent 100 reconstruction residuals (400 bytes in float32) and a few scalar calibration variables, requiring less than 1 kB of asset-specific state. The only fleet-wide artifact is the shared nominal representation, which is trained once and distributed to every operational unit.

Table 8: Edge-deployment footprint of the proposed shared nominal representation.

Metric	Value	Metric	Value
Model parameters	847,601	Residual-buffer memory	400 bytes
Raw float32 size	3.39 MB	Total calibration state	428 bytes (< 1 kB)
Serialized (.keras)	10.24 MB	Per-asset model training	None
CPU latency (median)	6.26 ms	Fault labels required	None
CPU latency (p95)	6.62 ms	Local residual buffer	100 samples

6 Mathematical Scope and Deployment Guarantees

This section states, without full proof, the minimal functional-analytic scaffolding justifying why one shared canonical representation and independently estimated local operational radii are mathematically compatible; complete derivations are given in our companion technical reports Di Santi 2026a; Di Santi 2026b.

> **Scope.** Jointly recovering a nuisance-invariant representation *and* calibrating per-asset decision boundaries across a heterogeneous fleet is combinatorially hard. This paper decouples that problem into (i) a *global* inverse problem, solved once offline, recovering a canonical representation whose nominal orbit is relatively compact (Proposition 1); and (ii) a *local* inverse problem, solved independently by each asset as an $O(1)$-per-observation empirical-quantile estimate.

6.1 Observation model

Let $X \subset \mathbb{R}^n$ denote observation windows, $\theta \in \Theta$ the latent diagnostic state, and $\eta \in E$ nuisance variables (sensor coupling, mounting, operating point, noise), with $x = F(\theta, \eta)$. The group G_{nuis} acts on E while preserving θ; observations are diagnostically equivalent, $x \sim x'$, iff $x' = F(\theta, g \cdot \eta)$ for some $g \in G_{\text{nuis}}$.

6.2 Compactness and factorization

Proposition 1 (Relative compactness of the nominal orbit)**.** *If E is compact and $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous on E, then $O_N = \{F(\theta_N, \eta) : \eta \in E\}$ is relatively compact in $(C(X), \|\cdot\|_\infty)$.*

Proof sketch. By Arzelà–Ascoli Rudin 1976, boundedness and equicontinuity on compact E give relative compactness of O_N; the equicontinuity modulus for the Stage 1 simulator is verified in Di Santi 2026b. □

Because the nominal orbit is compact, $\tau_k^\star$ only accounts for asset-specific scale around a fixed structure rather than searching an unbounded space, explaining the rapid convergence observed empirically (51.0 ± 12.6 windows on CWRU, Section 5.4).

Proposition 2 (Factorization through the nuisance quotient)**.** *Let $\pi : X \to X/G_{nuis}$ be the quotient projection. If $d : X \to \mathbb{R}$ is G_{nuis}-invariant, then d is constant on fibers and factors uniquely as $d = \bar{d} \circ \pi$.*

Proof sketch. Invariance makes d constant on fibers, so $\bar{d}([x]) := d(x)$ is well defined by the universal property of the quotient map Di Santi 2026b. □

Together, these results justify treating diagnosis as membership testing around a fixed canonical representation: $\bar{d}$ (equivalently O_N) is recovered once, globally, while each asset k solves only a well-posed local calibration, estimating $\tau_k^\star$ for coverage $1 - \alpha$ (Eq. (4)) from its own residuals. Empirically, this is reflected in the same fixed representation transferring across four CWRU conditions with only $\tau_k(t)$ re-identified (99.69% average ball-fault, 100% inner/outer-race detection), and extending to physically distinct NASA IMS bearings with 83.25 h average lead time.

7 Discussion

The principal result of this work is not that a Conv1D autoencoder can separate CWRU fault classes, a capability already demonstrated extensively in the literature. The contribution lies instead in the deployment methodology: a single canonical nominal representation can be shared across heterogeneous assets, while each physical unit independently estimates its own operational radius without retraining the representation. From a procurement perspective, this separation gives fleet operators a concrete basis for specifying deployment requirements: rather than accepting per-asset models developed ad hoc by each vendor, an operator can require a single validated, versioned nominal representation together with documented local calibration behaviour, independent of which organization, component OEM, third-party analytics provider, or in-house engineering team, produces and maintains it.

The repeated-run and initialization-robustness experiments jointly support this claim from two sides. Independent random initializations converge to slightly different reconstruction-error scales, yet healthy false-alarm rate and fault detection remain essentially unchanged after empirical calibration; conversely, keeping the representation fixed while varying only the 100-window warm-up sample produces only modest variation in the estimated operational radius. The deployment mechanism is therefore robust both in the learned representation and in the local calibration procedure. This confirms that the transferable object is not an absolute reconstruction-error threshold but a shared canonical representation together with a target nominal coverage: the calibration-transfer experiment shows the representation transferring successfully from simulation to field, while the absolute residual scale does not.

Identifying the shared representation as the transferable object also changes the engineering organization of fleet deployment. Rather than developing and maintaining an independent representation model for each physical asset, these activities can be centralized, for example by the original equipment manufacturer (OEM), while each deployed asset performs only a lightweight local calibration during commissioning. Computational scalability is correspondingly decoupled from fleet size: growth increases the number of monitored assets, not the number of engineered models, since adding a unit requires only distributing the shared representation and estimating its local radius. This deployment methodology also fits within the broader theoretical framework developed in our previous work, which formalizes industrial diagnosis as an inverse problem over quotient spaces Di Santi 2026b; Di Santi 2026a; Di Santi 2025; the present paper addresses only the deployment stage of that formulation.

Operational-radius convergence also clarifies the role of adaptation: its purpose is to identify the asset during cold start, not to track its aging indefinitely. Once $\tau_k^\star$ is established, continued movement of the anomaly score constitutes diagnostic evidence rather than input for further adaptation, and initialization could instead be triggered adaptively until a stability criterion is met, after which the radius is frozen or tightly constrained. Together with the edge footprint reported in Section 5.7, this deployment model enables day-one monitoring without fault labels, isolates asset-specific adaptation to local calibration, and requires only lightweight edge computation.

7.1 Limitations

Several limitations qualify the scope of the present results and should guide their interpretation.

Nominal initialization assumption. The procedure assumes the cold-start observations are representative of healthy operation. If an asset were already degraded at installation, $\tau_k^\star$ could absorb part of that condition into the nominal baseline; a consistency check against the canonical nominal manifold is a natural extension and is left as future work.

Short-warm-up sensitivity. Because the operational radius is an empirical 98th-percentile estimated from only 100 windows, it can be sensitive to a small number of extreme values, as observed in the 1797 RPM condition (5.91% FAR, roughly twice the other conditions). Longer warm-ups or robust quantile estimators would likely reduce this variability at the cost of a longer cold start.

Stationarity of nominal operation. The convergence criterion assumes an approximately stationary healthy residual distribution during cold start. The retrospective trajectories in Figure 4 show modest post-calibration drift; more variable startup conditions were not evaluated and may benefit from a longer or adaptively triggered calibration window.

Validation domain. The validation uses bearing datasets (CWRU, NASA IMS) rather than instrumented transportation assets. While the deployment principle is architecture- and modality-agnostic by construction (Section 3), its extension to other components remains to be confirmed empirically.

7.2 Future Work

A natural extension relaxes the assumption that assets begin in nominal condition. Because the methodology separates the shared representation from the local radius, future work will investigate validating the initialization phase itself against the canonical nominal manifold, to distinguish genuinely healthy startup from incipient degradation before it is absorbed into the baseline. A second direction concerns progressive degradation: whether the joint evolution of reconstruction residuals and operational-radius dynamics carries predictive information, connecting this methodology with physics-informed health indicators and remaining-useful-life estimation.

Finally, the mathematical interpretation in Di Santi 2026a suggests formalizing the operational radius as an asset-specific neighborhood around the transferable canonical representation, characterizing its invariance properties and the conditions under which local calibration preserves the underlying quotient-space structure.

8 Conclusion

We presented a day-one deployment methodology for nominal condition monitoring across heterogeneous asset fleets, validated on public rolling-element bearing datasets. A single nominal representation is trained once on simulated healthy observations and transferred unchanged to field deployment. Each operational unit then performs only a short unlabeled warm-up to estimate its own operational radius, without modifying or retraining the shared representation.

The experimental results show that a simulator-trained nominal representation transfers successfully to field deployment, whereas an absolute decision threshold does not. A shared target nominal coverage, realized locally through the empirical residual quantile estimated during initialization, provides an effective bridge between the shared representation and field deployment. This deployment strategy preserves detection performance close to that of asset-specific models trained under ideal asset-specific calibration conditions while eliminating per-asset representation learning.

The transferable object is therefore not an absolute anomaly threshold, nor an asset-specific representation, but a shared canonical nominal representation together with a lightweight local calibration mechanism. The central outcome of this work is consequently not a new anomaly detector, but a scalable deployment principle: learn the canonical nominal representation once, distribute it across the fleet, and let each physical asset identify only its own operational radius during installation.

This separation between representation learning and deployment calibration provides a practical foundation for large-scale edge condition monitoring, where an OEM can centrally produce and maintain the shared representation while each deployed asset performs its own lightweight local calibration automatically, without operator intervention or manual threshold tuning.

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